

Legal Analysis

<https://github.com/amberfqy/compstats-group-analysis.git>

Question: Does the a large monetary penalty of a legal case indicate a use of AI tool?

(0) Data Wrangling

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.6
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.1      v tibble     3.3.1
v lubridate  1.9.4      v tidyr      1.3.2
v purrr      1.2.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(dplyr)
df <- read_csv("Charlotin-hallucination_cases.csv")
```

Rows: 2098 Columns: 18

```
-- Column specification -----
Delimiter: ","
chr   (17): Case Name, Court, State(s), Party(ies), AI Tool, Hallucination It...
date  (1): Date
```

```
i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
df <- df |>
  mutate(
    Monetary_Penalty_num = parse_number(`Monetary Penalty`)
  ) |> filter(!is.na(Monetary_Penalty_num))

df <- df |>
  mutate(
    Binary_AI_Tool = case_when(
      str_to_lower(`AI Tool`) == "unidentified" ~ 0,
      !is.na(`AI Tool`) ~ 1,
      TRUE ~ NA_real_
    )
  ) |> filter(!is.na(Binary_AI_Tool))

head(df)
```

```
# A tibble: 6 x 20
  `Case Name`      Court `State(s)` Date      `Party(ies)` `AI Tool`
  <chr>            <chr> <chr>    <date>    <chr>        <chr>
1 State v. Sandoval / In re ~ SC N~ USA      NA        Lawyer      ChatGPT
2 Aguilar v. The Crawford Gr~ D. M~ USA    2026-09-25 Lawyer      Unidenti~
3 Englert v. Gateway Crossin~ CA I~ USA    2026-09-18 Pro Se Liti~ Implied
4 Tiekert v. Village of Mama~ S.D.~ USA    2026-09-14 Lawyer      ChatGPT
5 Dey v Minister for Immigra~ Fede~ Australia 2026-09-14 Lawyer      Unidenti~
6 Virgil v. Experian Informa~ S.D.~ USA    2026-09-11 Lawyer      Implied
# i 14 more variables: `Hallucination Items` <chr>, Outcome <chr>,
#   `Monetary Penalty` <chr>, `Professional Sanction` <chr>, Alleged <chr>,
#   `Vendor Disputed` <chr>, `Vendor Disputed Explanation` <chr>,
#   Pointer <chr>, Source <chr>, Details <chr>, `Legal Field Primary` <chr>,
#   `Legal Field Secondary` <chr>, Monetary_Penalty_num <dbl>,
#   Binary_AI_Tool <dbl>
```

(a) Statistical Model

```
m_logistic <- glm(Binary_AI_Tool ~ Monetary_Penalty_num, data = df, family = binomial)
summary(m_logistic)
```

Call:

```
glm(formula = Binary_AI_Tool ~ Monetary_Penalty_num, family = binomial,
     data = df)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	5.233e-01	1.182e-01	4.428	9.53e-06 ***
Monetary_Penalty_num	1.091e-05	8.562e-06	1.274	0.203

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

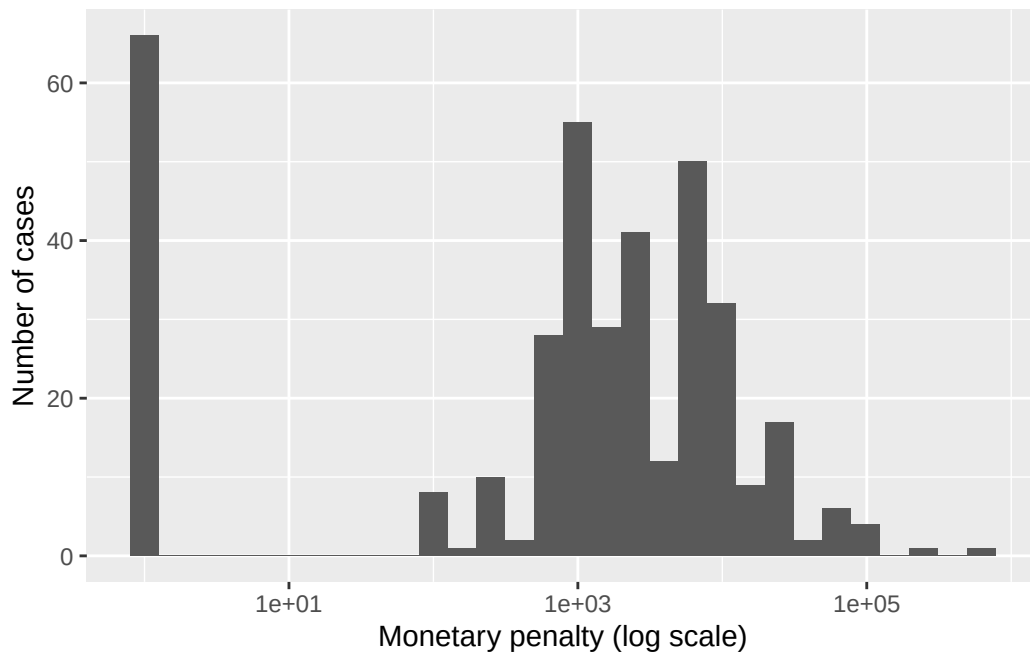
(Dispersion parameter for binomial family taken to be 1)

Null deviance: 486.84 on 373 degrees of freedom
Residual deviance: 483.78 on 372 degrees of freedom
AIC: 487.78

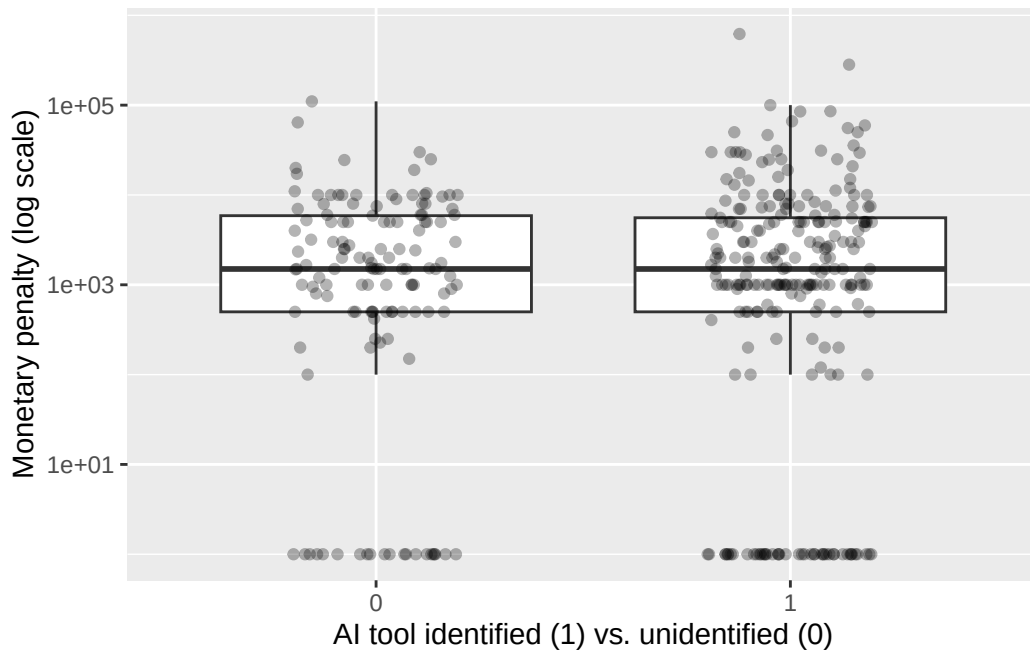
Number of Fisher Scoring iterations: 5

(b) Three Visualization

```
ggplot(data = df, aes(x = Monetary_Penalty_num)) +  
  geom_histogram(bins = 30) +  
  scale_x_log10() +  
  labs(x = "Monetary penalty (log scale)", y = "Number of cases")
```

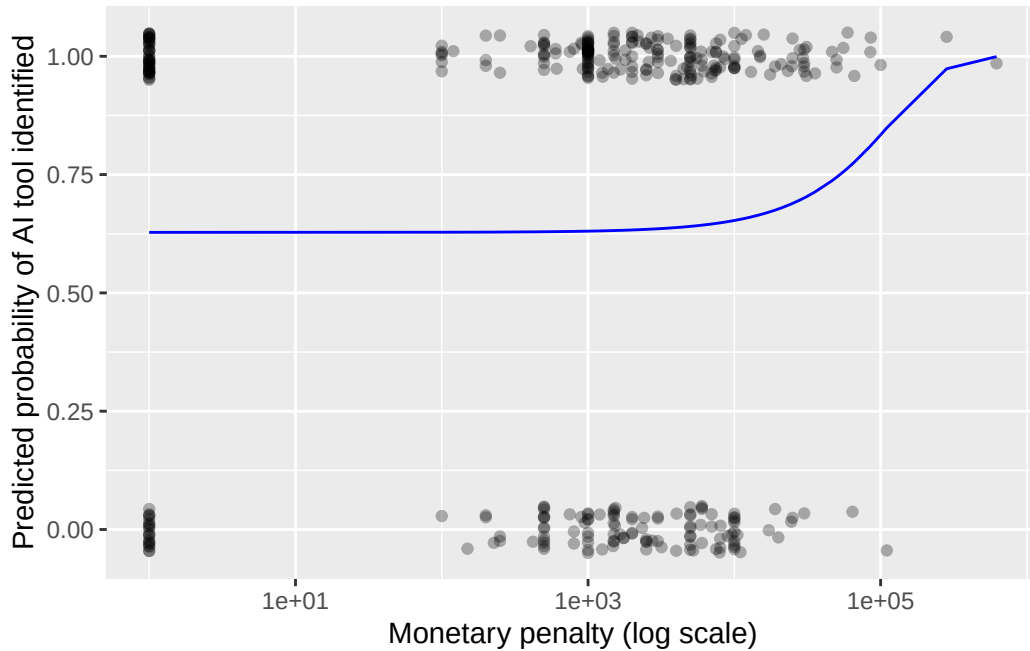


```
ggplot(data = df, aes(x = factor(Binary_AI_Tool), y = Monetary_Penalty_num)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter(width = 0.2, alpha = 0.3) +
  scale_y_log10() +
  labs(x = "AI tool identified (1) vs. unidentified (0)",
       y = "Monetary penalty (log scale)")
```



```
df <- df |>
  mutate(prob = predict(m_logistic, type = "response"))

ggplot(data = df, aes(x = Monetary_Penalty_num)) +
  geom_jitter(aes(y = Binary_AI_Tool), height = 0.05, alpha = 0.3) +
  geom_line(aes(y = prob), color = "blue") +
  scale_x_log10() +
  labs(x = "Monetary penalty (log scale)",
       y = "Predicted probability of AI tool identified")
```



(c) Why the model is appropriate

Our response variable, `Binary_AI_Tool`, only takes the values 0 and 1, so linear regression is not appropriate because it can give predicted values not within 0-1. Logistic regression models the log-odds of the response as a linear function of the predictor, so the predicted probabilities always stay between 0 and 1. This lets us check whether a larger monetary penalty is associated with a higher probability that the AI tool was identified.

(d) Interpretation of the model results

```
exp(10000 * coef(m_logistic)["Monetary_Penalty_num"])
```

```
Monetary_Penalty_num
1.11528
```

The p-value for the monetary penalty is 0.203, which is larger than 0.05. Each \$10,000 increase in the penalty multiplies the odds that the AI tool was identified by about 1.115, but this effect is not statistically significant. So we do not have evidence that a larger penalty is related to the AI tool being identified. This matches the boxplot, where both groups have similar penalties.

(e) Limitations

This analysis on legal case and AI hallucination still have several limitations: 1) We try to use monetary penalty as a predictor for whether AI tool is used in the case or not. However, by common sense, this is not the best predictor for AI tool, which is confirmed again by our model. A better predictor would be a qualitative analysis of the textual evidence presented in the court and interpret through whether AI is used or not. However, due to the data set limitation, the quantification of such a predictor is hard to achieve. 2) We only kept cases that had a recorded monetary penalty, which leaves 374 cases out of the full database. Most cases ended with a warning or another sanction instead of a fine, so our results may not apply to those cases.